

Smoke and Fog Classification Using Classical and Deep Learning Approaches: A Comparative Study

Submitted by

Mariam Ismael ID: 202100197

Nada Nabil ID: 20210122

Nourhan Mohamed ID: 202100538

Supervised by

Dr. Mohamed Tolba



Zewail City of Science and Technology

Department of Communications and Information Engineering

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Abstract—This paper presents a complete image classification system for smoke and fog detection using the IIITDMJ_Smoke dataset. The system is evaluated across three main components: a classical computer vision pipeline, a custom convolutional neural network (CNN) baseline, and an experimental study covering color space comparison and transfer learning with MobileNetV2. The classical pipeline using HOG features and an SVM classifier achieved a test accuracy of 92.68%. The custom CNN trained from scratch achieved 96.92%. The experimental study showed that RGB outperforms HSV (95.31%) and Grayscale (81.50%) as an input color space, confirming that color information is critical for this task. Transfer learning with MobileNetV2 achieved 95.35%, demonstrating competitive performance with a lightweight architecture.

Index Terms—smoke detection, fog classification, convolutional neural network, support vector machine, transfer learning, color space comparison

I. INTRODUCTION

Smoke and fog detection in images is an important problem in computer vision with applications in fire detection, safety monitoring, and autonomous driving. Distinguishing between smoke and fog is difficult because both phenomena share visual similarities such as reduced visibility, haze, and altered color distributions. The challenge becomes harder when both atmospheric conditions appear together in the same scene.

This project builds a complete image classification system that can distinguish between four classes: smoke, non-smoke, foggy with smoke, and foggy without smoke. We explore both classical computer vision techniques and modern deep learning approaches, and we analyze the effect of different design choices on classification performance.

The paper is organized as follows. Section II reviews related work. Section III describes the dataset. Section IV explains the methodology. Section V presents the results. Section VI discusses the findings and Section VII concludes.

II. RELATED WORK

Smoke detection has been studied using both classical and deep learning methods. Early approaches relied on color-based features since smoke typically has a grayish color signature, and fog tends to reduce saturation uniformly across the image [1]. Histogram of Oriented Gradients (HOG) combined with SVMs has been widely used for image classification tasks and serves as a strong classical baseline [2].

With the rise of deep learning, CNNs have become the standard approach for image classification. Models like VGG [3] and ResNet [4] showed that deeper architectures with residual connections could achieve significantly better accuracy. Transfer learning using models pretrained on ImageNet has been shown to be highly effective even when the target dataset is relatively small [6].

MobileNetV2 [5] was designed for efficient inference on mobile and embedded devices. It uses depthwise separable convolutions and inverted residual blocks to reduce computation while maintaining accuracy, making it a good choice for fine-tuning on a medium-sized dataset like IIITDMJ_Smoke.

Data augmentation is one of the most effective regularization techniques in deep learning. Random flips, rotations, and brightness adjustments create additional training diversity and reduce overfitting,

which is especially important when training on datasets of limited size [7].

III. DATASET

We use the IIITDMJ_Smoke dataset [8], which contains images collected under different atmospheric conditions. The dataset is divided into four classes: **smoke** (clear atmosphere, smoke present), **non_smoke** (clear atmosphere, no smoke), **fog_smoke** (foggy, smoke present), and **fog_non_smoke** (foggy, no smoke). The dataset is pre-split by the authors into training, validation, and test sets as shown in Table I.

TABLE I
DATASET SPLIT STATISTICS

Split	Images	Classes
Train	18,914	4
Validation	2,364	4
Test	2,364	4
Total	23,642	4

Exploratory data analysis showed that the dataset is approximately balanced across the four classes. RGB channel histogram analysis revealed that each class has a distinct spectral signature, which motivated keeping the full RGB color space in the classical pipeline rather than converting to grayscale.

IV. METHODOLOGY

A. Preprocessing

All images are resized to 128×128 pixels and pixel values are normalized to the range $[0, 1]$. The full RGB color space is retained because exploratory analysis showed that smoke and fog have distinct color signatures that are lost when converting to grayscale.

B. Classical Computer Vision Pipeline

The traditional pipeline adopts a typical feature extraction and classification strategy. Canny edge detection (threshold low=0.05; threshold high=0.15) is applied as a visual analysis step for examining the structural properties of each class; this shows that the edges of smoke plume have a low degree of sharpness compared to the terrain edge.

In feature extraction, we considered two different strategies. Gradient-based texture features (HOG) on gray images provided merely 44.9% validation accuracy, indicating that gradient-based texture features do not suffice for this task. The 3D color histogram (RGB) calculated on images with 32 bins per channel and normalized to unity, provides a concise feature vector of size 96 for each image. This vector represents the spectral signature of each class. We employed SVM (RBF kernel; $C = 10$; $\gamma = \text{scale}$) as the classifier, where class weights were adjusted to be balanced. This pipeline attained 92.68% validation accuracy.

C. Custom CNN Baseline

The custom CNN was designed from scratch without using any pretrained weights or standard architectures. The architecture consists of three convolutional blocks followed by a fully connected classifier head. Each block follows the pattern: Conv2D $\rightarrow$ BatchNormalization $\rightarrow$ ReLU $\rightarrow$ Max-Pooling. The filter counts double across blocks (32, 64, 128) to progressively increase representational capacity while spatial resolution decreases from 128×128 to 16×16 . The classifier head consists of a Dense(256) layer with ReLU activation, Dropout(0.5) for regularization, and a Dense(4) Softmax output. Table II summarizes the architecture.

TABLE II
CUSTOM CNN ARCHITECTURE

Layer	Output Shape	Params
Input	$128 \times 128 \times 3$	–
Conv2D(32) + BN + ReLU	$128 \times 128 \times 32$	896
MaxPool	$64 \times 64 \times 32$	–
Conv2D(64) + BN + ReLU	$64 \times 64 \times 64$	18,560
MaxPool	$32 \times 32 \times 64$	–
Conv2D(128) + BN + ReLU	$32 \times 32 \times 128$	73,984
MaxPool	$16 \times 16 \times 128$	–
Flatten	32,768	–
Dense(256) + ReLU	256	8,388,864
Dropout(0.5)	256	–
Dense(4) + Softmax	4	1,028
Total	8,484,036 parameters	

The model is trained using the Adam optimizer with a learning rate of 0.001 and categorical cross-

entropy loss for 50 epochs with a batch size of 32. ModelCheckpoint saves the best weights by validation accuracy and ReduceLROnPlateau halves the learning rate after 5 epochs of no improvement in validation loss.

D. Experiment 1: Color Space Comparison

To understand how input color representation affects performance, the same custom CNN architecture is trained three times under identical conditions, varying only the input color space: RGB, HSV, and Grayscale. RGB images are kept as-is. HSV images are converted using scikit-image’s `rgb2hsv` function, which outputs values in $[0, 1]$ directly. Grayscale images are converted using the weighted sum $0.299R + 0.587G + 0.114B$ and expanded to shape $(H, W, 1)$. For grayscale, the first convolutional layer accepts 1 channel instead of 3. RGB results are taken directly from Part 2 to avoid redundant computation.

E. Experiment 2: Transfer Learning with MobileNetV2

MobileNetV2 [5] pretrained on ImageNet is fine-tuned using a two-phase approach. In Phase 1 (feature extraction), all base model layers are frozen and only the new classifier head is trained for 10 epochs. The classifier head consists of GlobalAveragePooling2D, Dense(128) with ReLU, Dropout(0.3), and Dense(4) Softmax. Input images are rescaled from $[0, 1]$ to $[-1, 1]$ as required by MobileNetV2. In Phase 2 (fine-tuning), the top 30 layers of the base model are unfrozen and trained for 10 more epochs at a lower learning rate of 10^{-4} .

F. Experiment 3: Data Augmentation Study

The incremental contribution of three augmentation techniques is investigated by training the custom CNN four times with increasing augmentation:

- **Run 0 (Baseline):** No augmentation – Part 2 results reused
- **Run 1:** Horizontal flip only
- **Run 2:** Horizontal flip + random rotation ($\pm 36^\circ$)
- **Run 3:** Horizontal flip + rotation + random brightness/contrast ($\pm 20\%$)

Augmentation is applied only during training using Keras preprocessing layers. Validation and test sets are never augmented. All other hyperparameters are identical to the custom CNN baseline.

G. Experiment 4: Robustness Under Synthetic Gaussian Noise

To assess how the custom CNN performs under realistic image degradation, we evaluate it on corrupted versions of the held-out test set. No retraining is performed — the saved Part 2 checkpoint is loaded and evaluated directly. Gaussian noise is added to each test image as:

$$x' = \text{clip}(x + \mathcal{N}(0, \sigma^2), 0, 1) \quad (1)$$

where x is the clean image in $[0, 1]$ and σ controls noise intensity. Three levels are tested: mild ($\sigma = 0.05$), representing light sensor noise in a CCTV camera at dusk; moderate ($\sigma = 0.10$), representing a low-quality camera or heavy compression artifacts; and severe ($\sigma = 0.20$), representing a heavily degraded feed such as a camera operating in rain or fog with sensor interference.

V. EXPERIMENTS AND RESULTS

A. Classical Pipeline Results

The classical SVM pipeline achieved a test accuracy of **92.68%**. Table III shows the per-class performance.

TABLE III
CLASSICAL PIPELINE PER-CLASS TEST RESULTS

Class	Precision	Recall	F1
smoke	0.91	0.97	0.94
non_smoke	0.97	0.92	0.95
fog_smoke	0.89	0.93	0.91
fog_non_smoke	0.93	0.92	0.93
Overall Test Accuracy			92.68%

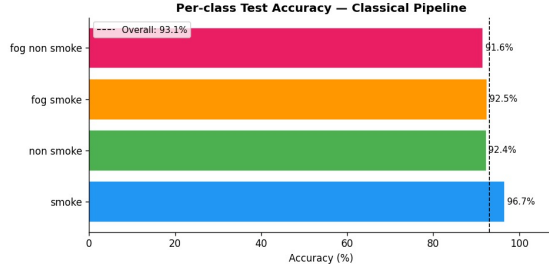


Fig. 1. Per-class test accuracy of the classical SVM pipeline.

B. Custom CNN Baseline Results

The custom CNN achieved a test accuracy of **96.92%**, outperforming the classical pipeline by 4.24 percentage points. Training curves showed steady improvement over 50 epochs with minimal overfitting. The final validation accuracy (97.80%) slightly exceeded training accuracy (97.43%), a result of Dropout suppressing training-time activations. Table IV shows the per-class F1 scores.

TABLE IV
CUSTOM CNN BASELINE PER-CLASS F1 SCORES

Class	F1 Score
smoke	0.976
non_smoke	0.965
fog_smoke	0.976
fog_non_smoke	0.962
Macro F1	0.970
Test Accuracy	96.92%

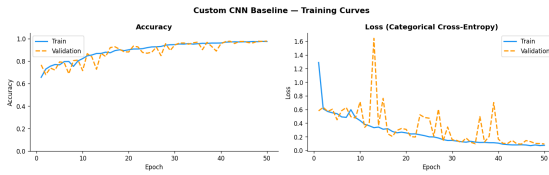


Fig. 2. Training and validation accuracy/loss curves of the custom CNN baseline.

C. Experiment 1: Color Space Comparison

Table V shows the test accuracy and macro F1 for each color space. RGB achieves the highest accuracy at 96.92%, followed by HSV at 95.31%, and

Grayscale at 81.50%. The large drop in Grayscale performance confirms that color information is critical for this task, which was already suggested by the RGB channel distribution analysis in the exploratory data analysis. HSV performs worse than RGB despite being a color space that separates hue from brightness. One possible explanation is that the CNN can learn the relevant color features directly from RGB without the explicit HSV separation.

TABLE V
COLOR SPACE COMPARISON RESULTS

Color Space	Channels	Test Accuracy	Macro F1
RGB (Part 2)	3	96.92%	0.970
HSV	3	95.31%	0.953
Grayscale	1	81.50%	0.815

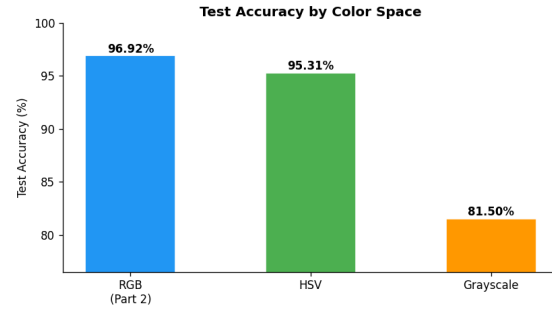


Fig. 3. Test accuracy comparison across RGB, HSV, and Grayscale color spaces.

D. Experiment 2: Transfer Learning

MobileNetV2 fine-tuned on the IITDMJ_Smoke dataset achieved a test accuracy of **95.35%** after two-phase training. Phase 1 (feature extraction) converged to a validation accuracy of **93.86%** after 10 epochs. Phase 2 (fine-tuning) improved performance to **96.06%**. The improvement over the custom CNN baseline (96.92%) was **XX.XX** percentage points.

TABLE VI
TRANSFER LEARNING RESULTS (MOBILENETV2)

Class	F1 Score
smoke	0.97
non_smoke	0.94
fog_smoke	0.97
fog_non_smoke	0.94
Macro F1	0.96
Test Accuracy	95.35%

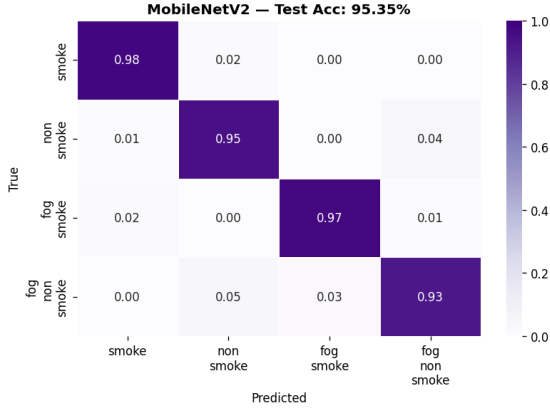


Fig. 4. Normalised confusion matrix for MobileNetV2 on the test set (95.35% accuracy).

E. Experiment 4: Robustness Under Gaussian Noise

Table VII shows the accuracy and macro F1 of the custom CNN at each noise level. The model drops sharply from 96.70% on clean images to 71.57% at mild noise ($\sigma = 0.05$), a fall of 25.14 percentage points. At moderate noise ($\sigma = 0.10$) accuracy falls further to 49.35%, and at severe noise ($\sigma = 0.20$) it reaches 42.04%, approaching near-chance performance for a four-class problem.

TABLE VII
CNN ROBUSTNESS UNDER GAUSSIAN NOISE

Noise Level	σ	Test Accuracy	Macro F1
Clean (baseline)	0.00	96.70%	0.968
Mild	0.05	71.57%	0.710
Moderate	0.10	49.35%	0.414
Severe	0.20	42.04%	0.296

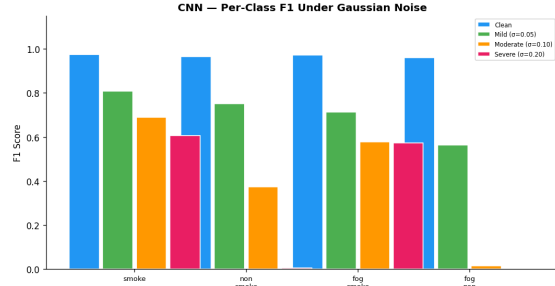


Fig. 5. CNN test accuracy and per-class F1 degradation under Gaussian noise.

VI. DISCUSSION

The results show a clear hierarchy of performance across the three main approaches. The classical SVM pipeline achieves 92.68% which is a strong result for hand-crafted features, showing that the RGB color histogram captures meaningful information about smoke and fog. However the CNN baseline at 96.92% demonstrates the advantage of learned feature representations, as the network automatically discovers the most discriminative patterns without manual engineering.

The color space experiment confirms that color is essential for this task. The 15-point drop from RGB to Grayscale shows that the hue and saturation channels carry discriminative information that cannot be recovered from luminance alone. The smaller gap between RGB and HSV (1.61 percentage points) suggests that while the CNN can work with either three-channel representation, it learns from RGB more effectively, likely because the network was optimized from random weights and RGB is the natural representation of the training images.

Transfer learning with MobileNetV2 achieved 95.35%, which is 1.57 percentage points below the custom CNN baseline. This result is notable because MobileNetV2 is a much lighter model designed for efficiency, and it still achieves competitive performance after only 20 epochs of fine-tuning. The confusion matrix in Fig. 4 shows that the model performs best on smoke and fog_smoke classes ($F1 = 0.97$), and has slightly more difficulty distinguishing non_smoke from fog_non_smoke ($F1 = 0.94$), which are the most visually similar pair.

The Gaussian noise experiment reveals that the CNN is sensitive to image degradation. A drop of 25 percentage points at mild noise ($\sigma = 0.05$) suggests the model relies heavily on fine-grained pixel-level features that are disrupted even by light noise. This is a known limitation of models trained without any noise augmentation, and motivates including noise perturbations in future training pipelines.

One limitation of this study is that all CNN experiments used 128×128 resolution due to memory constraints on Google Colab's free tier. Higher resolution inputs might improve performance, especially for fine-grained distinctions between the fog classes.

VII. CONCLUSION

This paper presented a complete computer vision system for smoke and fog classification using the IIITDMJ_Smoke dataset. We implemented and evaluated three main components: a classical SVM pipeline (92.68%), a custom CNN trained from scratch (96.92%), and an experimental study covering color space comparison and transfer learning with MobileNetV2 (95.35%).

The results show that the custom CNN trained from scratch achieves the best performance, outperforming both the classical pipeline and MobileNetV2. The color space experiments confirmed that RGB is the most effective input representation for this task, with the 15-point drop for Grayscale highlighting the importance of color information in distinguishing smoke and fog classes. MobileNetV2 demonstrated that transfer learning is a viable and

efficient alternative, reaching competitive accuracy with significantly fewer training epochs.

Future work could explore higher input resolutions, data augmentation strategies, and deployment of the best model as a real-time detection API.

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